SET 4: THE ADIRI ICE-RIDGE DEUTERIUM RECOVERY & CRYOGENIC HYDROGEN STORAGE DEPOT

DOCUMENT 4B: BORROWER'S Q3 2146 QUARTERLY REPORTING PACKAGE & VALUATION MEMO

To: Saturn Orbital Development Syndicate (SODS), as Administrative Agent

From: Adiri Fuel Reserves LP (Borrower)

Date: October 20, 2146

Subject: Q3 2146 Financial Compliance Report & Asset Valuation Analysis

1. Financial and Operational Narrative

During the third quarter of 2146, the Adiri Fuel Reserves facility operated in complete compliance with all covenants and technical metrics detailed in the Secured Credit Agreement (Document 4A). The depot's containment systems maintained high thermodynamic efficiency, achieving a quarterly average **Cryo-Storage Pressure Tolerance Index (CSPTI)** of 96.2%, which exceeds the 93.0% covenant standard specified in Section 4.01(1).

Deuterium recovery volumes averaged 1,285 kilograms of high-purity isotopic deuterium per Titan sidereal day, meeting the **Deuterium Extraction Yield Target** of 1,200 kilograms outlined in Section 4.01(2). Consequently, the Facility Loans accrued interest at the baseline interest rate of **TIDOR plus 5.75%** per annum, with no triggers of the Pressure Containment Risk Premium. In addition, the **Helium-Boiloff Sinking Reserve** balance was fully maintained at the required SC 300,000,000 under Section 4.01(3).

2. Financial Statements (SSIAS Compliant)

Table 4.1: Consolidated Balance Sheet & Income Statement (Q3 2146)

All figures in Millions of Sol Credits (SC)

BALANCE SHEET & INCOME STATEMENT CONSOLIDATION			
Category / Account Item	**Q3 2146**	**Q2 2146**	**Variance (%)**
ASSETS			
Cash & Liquid Cryo-Balances	385.20	360.10	+6.97%
Helium-Boiloff Sinking Reserve[1]	300.00	300.00	+0.00%

Refined Fuel & Deuterium Inventories	490.50	460.00	+6.63%
Depot Property, Plant & Equipment (Net)[2]	3,650.00	3,710.00	-1.62%
TOTAL ASSETS	**4,825.70**	**4,830.10**	**-0.09%**
LIABILITIES & PARTNERS' CAPITAL			
Secured Term Loans (SODS Facility)[3]	3,150.00	3,250.00	-3.08%
Seismic Anchoring Remediation Provision	520.00	515.00	+0.97%
Partners' Capital Accounts	1,155.70	1,065.10	+8.51%
TOTAL LIABILITIES & CAPITAL	**4,825.70**	**4,830.10**	**-0.09%**
OPERATING INCOME STATEMENT			
Orbital Deuterium Sales[4]	542.10	495.20	+9.47%
Cryo-Centrifuge Operating Expenses[5]	(245.30)	(241.00)	+1.78%
Net Operating Income (NOI)	**296.80**	**254.20**	**+16.76%**

[1] Reconciles directly to the mandatory cash reserve account specified in Section 4.01(3).

[2] Detailed asset class values are reconciled directly to the subtotal summation of Net Book Values reported in **Table 4.2**.

[3] Reconciles directly to the outstanding Facility Loan principal balance specified under Section 2.01.

[4] Annualized orbital deuterium sales correlate directly with the Base Case model outlined in **Table 4.3**.

[5] Includes costs associated with helium compression cycles, vacuum jacket maintenance, and structural seismic monitoring.

Table 4.2: Detailed Depot Property, Plant & Equipment (PP&E)

All figures in Millions of Sol Credits (SC)

Asset Category	Sub-Asset Class	Gross Cost	Accumulated Depr.	Net Book Value
1. Cryo-Containment Tanks	Liquid Hydrogen Storage Chambers	1,550.00	(220.00)	1,330.00
	Vacuum-Jacket MLI Insulation Assemblies	720.00	(140.00)	580.00
	Subtotal Containment Tanks	**2,270.00**	**(360.00)**	**1,910.00**
2. Isotopic Separators	Centrifugal Enrichment Columns	1,100.00	(260.00)	840.00
	Feedstock Regolith Melters	850.00	(210.00)	640.00
	Subtotal Centrifugal Separators	**1,950.00**	**(470.00)**	**1,480.00**
3. Docking Infrastructure	Orbital Transfer Gantry & Cables	410.00	(150.00)	260.00
	Subtotal Docking Assets	**410.00**	**(150.00)**	**260.00**
TOTAL DEPOT PP&E[*]		**4,630.00**	**(980.00)**	**3,650.00**

[*] Reconciles directly to the **Depot Property, Plant & Equipment (Net)** line item on the Balance Sheet in **Table 4.1**.

3. Manager's Valuation Memo

Valuation Methodology: The Adiri Ice-Ridge Fuel Depot is valued utilizing a Discounted Cash Flow (DCF) model reflecting specific cryogenic containment risk variables. Key structural discount assumptions include a Weighted Average Cost of Capital (WACC) of 12.25% and centrifuge column mechanical wear cycles over a 20-year projected operational lifespan.

Table 4.3: Net Asset Value (NAV) Modeling under Alternate Pricing Scenarios

All figures in Millions of Sol Credits (SC) unless specified

Valuation Parameter	Base Case (Deuterium @ SC 4,200/kg)	Bear Case (Deuterium @ SC 3,100/kg)	Bull Case (Deuterium @ SC 5,300/kg)
Annual Deuterium Yield (k kg)	516.29	480.00	550.00
Gross Revenue (SC/yr)[#]	2,168.40	1,488.00	2,915.00
Cryo-Compressor Utility Costs	(820.00)	(850.00)	(790.00)
Vacuum Jacket Replacements	(161.20)	(161.20)	(161.20)
Net Operating Income (NOI)	**1,187.20**	**476.80**	**1,963.80**
Capitalized Value (8.5% Capitalization Rate)	**13,967.06**	**5,609.41**	**23,103.53**
Debt Outstanding (SODS Facility)[^]	(3,150.00)	(3,150.00)	(3,150.00)
Implied Net Asset Value (NAV)	**10,817.06**	**2,459.41**	**19,953.53**

[#] Pro-rata Q3 Gross Revenue matches the **Orbital Deuterium Sales** reported in **Table 4.1** (SC 542.10m represents exactly 25.0% of the annualized Base Case Gross Revenue of SC 2,168.40m).
[^] Corresponds to the **Secured Term Loans (SODS Facility)** balance reported in the liabilities section of **Table 4.1**.